

Predicting rate of employee attrition using machine learning algorithms

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Abstract—Employees are the most valuable assets of the organization. Their presence adds value to the organization in terms of both quality and quantity. When employees leave the organization, they carry with them invaluable tacit knowledge which is often the source of competitive advantage for the business. Thus, business leaders sometimes have to face huge setbacks due to the loss of talented employees. Irrespective of the kind of industry or the structure of the business, attrition is a common problem in every organization which not only hampers production but also results in heavy long-run costs and loss of goodwill to the organization. In this paper, an attempt has been made to analyse the publicly available employee attrition dataset using suitable models, study them and propose the best model on the basis of their prediction accuracies of employee attrition.

Index Terms—attrition, machine learning, prediction

I. INTRODUCTION

Attrition is the departure of employees from the organization for any reason (voluntary or involuntary), including resignation, termination, death or retirement. It is an inevitable part of any business. Some forms of attrition are unavoidable, like if an employee is retiring or is moving to another city. But after a certain threshold, attrition can make a big dent in a company's bottom line as well as its culture. Over the years, maintaining a permanent and promising workforce has become a tough task for employers and thereby increased attrition in organizations. Therefore, attrition is an alarming concern for all organizations across various sectors. Anything that involves manpower would be affected by employee attrition. Therefore nowadays, the importance and relevance of the employees are well understood by organizations and hence the focus is on understanding the key factors that affect employee attrition. The prediction of employee attrition at multinational companies can be very crucial for its overall growth.

Therefore, we have applied various machine learning and deep learning algorithms to predict the rate of employee attri-

tion. The aim of this research paper is to analyse the employee attrition dataset using various machine learning models and algorithms like Logistic Regression, Support Vector Machine (SVM) and Random Forest Model. We have compared the accuracies and F1 scores using graphs, and have chosen the best model out of these, for the prediction of employee attrition.

II. LITERATURE SURVEY

Middle level officers are more likely to leave, may be due to some disagreement with their senior officer as proposed by [3]. They observed major factors that influenced employee abandonment from the firm. The two rules are moderately derived by him. Some set of questions are asked with the both parties and depending upon their answers he concluded some facts based on workload, objectives, carrier opportunity and firm management. Human resource management [4] endeavors on basically termination rates and dismissal rates but actual content of them are enormously different. The previous model shows that, there are several distinct levels of attrition and turnover. Some research dictates that the consequences of dismissal and termination rates are at organizational level. Allen and Meyer (1990) [5] described the three-basic entity for the negative side of the turnover.

Regulating officer will more probable leave from the organization because of a contention with the higher administration than a representative who is in struggle with his prompt director. He recognized the determinant figures that influence employee acceptance [5] without protest from the organization. Two arrangements of information social occasion techniques were directed. An equivalent number of representative and officer respondents were solicited to answer a set from polls that were ordered by workload, objectives, identity, professional success, and hierarchical administration. The after-effects of the two information gathering methods demonstrated that the most noteworthy component that adds

to employee rejection [6] is money related compensation.

Many types of research have been made on the given topic using different algorithms, which are explored below. The given algorithms get updated over time with the change in parameters contributing to the decision of employees to resign. For instance, [1] uses various models such as SVM, Decision Trees, random forest, and KNN on the same IBM dataset. It evaluates the following by plotting visualization graphs of gender, travel, overtime vs Attrition. Based on the given values, it predicts the employee turnover which turns out to be the greatest in Random Forest with an accuracy of 88.43 percent in comparison to other classifiers.

In [2], the paper predicts the accuracy of five base models and later combines them to have a stronger predictor model which is often known as ensemble learning. The paper combines decision trees with linear regression to get an accuracy of 86.39 percent among others such as Adaboost and Random Forest and third as SVM and gradient boosting.

In [7], the paper works on a dataset available on Kaggle and works on well-known classification techniques, namely, Logistic Regression, Support Vector Machine (SVM), Random Forest, Decision Tree, AdaBoost, and Neural Network. It focuses on 12 main attributes uses three methods to derive accuracy that is Brute Force method, One Hot Encoding, and the Feature selection approach. The accuracy achieved is highest in the Feature Selection Approach with algorithm Random Forest which is a good accuracy achieved. Therefore, the authors of the paper decided to explore the results of ensemble trees like Random Forest and Gradient Boosting by using a feature selection approach.

III. METHODOLOGY

A. Dataset

For our project, we have taken a sample employee attrition dataset from kaggle. The dataset contains 1470 records and 35 fields including categorical and numeric features. Each record in the employee data set represents a single employee information and each field in the record represents a feature of that particular employee.

B. Pre-processing of dataset

Data pre-processing is one of the most important aspects of machine learning, and estimated to cover about 60% of the time and energy spent on a data science project. Therefore, it should be carried out with most sincerity and attention, as it forms the backbone of the further stages of the project process.

1) *Data Selection*: Firstly, Data Selection is done: data relevant to the project is selected from the initial dataset. For example, characteristics like ‘EmployeeCount’, ‘EmployeeNumber’, ‘Over18’, ‘StandardHours’ etc are removed, which are insignificant and redundant. Then, Binary Encoding, Ordinal Encoding and One-hot encoding

	mean	std	min	25%	50%	75%	max
Age	36.923810	9.135373	18.000000	30.000000	36.000000	43.000000	60.000000
DailyRate	802.485714	403.509100	102.000000	465.000000	802.000000	1157.000000	1499.000000
DistanceFromHome	9.192517	8.106864	1.000000	2.000000	7.000000	14.000000	29.000000
	mean	std	min	25%	50%	75%	max
Education	2.912925	1.024165	1.000000	2.000000	3.000000	4.000000	5.000000
EmployeeCount	1.000000	0.000000	1.000000	1.000000	1.000000	1.000000	1.000000
EmployeeNumber	1024.865306	602.024335	1.000000	491.250000	1020.500000	1555.750000	2068.000000
EnvironmentSatisfaction	2.721769	1.093082	1.000000	2.000000	3.000000	4.000000	4.000000
HourlyRate	65.891156	20.329428	30.000000	48.000000	66.000000	83.750000	100.000000
JobInvolvement	2.729932	0.711561	1.000000	2.000000	3.000000	3.000000	4.000000
JobLevel	2.063946	1.106940	1.000000	1.000000	2.000000	3.000000	5.000000
JobSatisfaction	2.728571	1.102846	1.000000	2.000000	3.000000	4.000000	4.000000
MonthlyIncome	6502.931293	4707.956783	1009.000000	2911.000000	4919.000000	8379.000000	19999.000000
MonthlyRate	14313.103401	7117.786044	2094.000000	8047.000000	14235.500000	20461.500000	26999.000000
NumCompaniesWorked	2.693197	2.498009	0.000000	1.000000	2.000000	4.000000	9.000000
PercentSalaryHike	15.209524	3.659938	11.000000	12.000000	14.000000	18.000000	25.000000
PerformanceRating	3.153741	0.360824	3.000000	3.000000	3.000000	3.000000	4.000000
RelationshipSatisfaction	2.712245	1.081209	1.000000	2.000000	3.000000	4.000000	4.000000
StandardHours	80.000000	0.000000	80.000000	80.000000	80.000000	80.000000	80.000000
StockOptionLevel	0.793878	0.852077	0.000000	0.000000	1.000000	1.000000	3.000000
TotalWorkingYears	11.279592	7.780782	0.000000	6.000000	10.000000	15.000000	40.000000
TrainingTimesLastYear	2.799320	1.289271	0.000000	2.000000	3.000000	3.000000	6.000000
WorkLifeBalance	2.761224	0.706476	1.000000	2.000000	3.000000	3.000000	4.000000
YearsAtCompany	7.008163	6.126525	0.000000	3.000000	5.000000	9.000000	40.000000
YearsInCurrentRole	4.229252	3.623137	0.000000	2.000000	3.000000	7.000000	18.000000
YearsSinceLastPromotion	2.187755	3.222430	0.000000	0.000000	1.000000	3.000000	15.000000
YearsWithCurrManager	4.123129	3.568136	0.000000	2.000000	3.000000	7.000000	17.000000

Fig. 1. Data Exploration figure

is carried out to transform the qualitative variables into quantitative variables and the categorical data to numbers so that the machine learning model could work. The original dataset in fact contained several variables with textual values (“BusinessTravel”, “Department”, “EducationField”). Therefore we applied a transcoding to transform the n values of a class into numeric variables, from 0 to n minus 1. For example, the “BusinessTravel” variable, consisting of 3 values, was transformed as follows:

- The “Non-Travel” value was mapped with the value 0,
- The “Travel-Frequently” value was mapped with the value 1,
- The “Travel-Rarely” value was mapped with the value 2.

2) *Data Exploration*: In order to observe the characteristics of all the variables, we considered the following variables: count, unique, top, frequency, mean, standard deviation (std), minimum and maximum values (min/max), 25%/50%/75% percentile. An extract of the overall dataset is reported in Fig.1 .

In Fig.2 , we report the correlation matrix heat map. It graphically illustrates correlations among all variables: the bluish fields represent no correlation, while the relative intensity of the dark blue and white colours represents an increase in correlation. In particular, dark blue reveals a positive or direct correlation (the variation of one characteristic directly affects the other) and white reveals a negative or indirect correlation (the variation of a characteristic inversely affects that of the other).

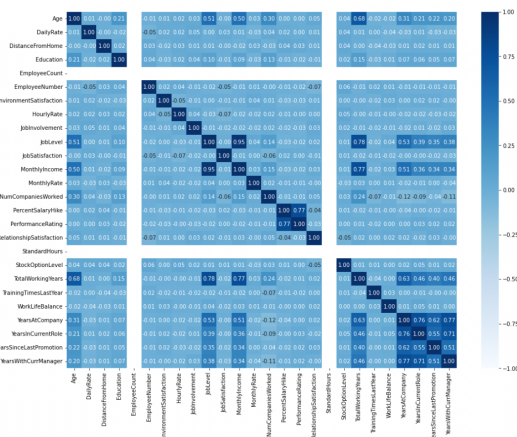


Fig. 2. Correlation matrix figure

3) *Test dataset and training dataset:* Separating data into test datasets and training datasets is an important part of evaluating data mining models. The test data set contains all the required data for data prediction and training data set contains all irrelevant data. Here we have 441 records in test dataset and 1029 records in training dataset. We apply data classification and data prediction on the test dataset of 441 records.

C. Model Building

The modelling process consists of selecting the best models, based on various machine learning techniques. In this case various predictive models were used such as those based on random forest, logistic regression and SVM. The goal is to identify the best classifier for the analysed problem. Each classifier must therefore be trained on the featured set and the classifier with the best classification results is used for prediction. The classification algorithms taken into consideration are:

- (i) Logistic Regression classifier,
- (ii) Support Vector Machine classifier,
- (iii) Random Forest classifier.

Random Forest Classifier: The random forest is a classification algorithm consisting of many decisions trees. It uses bagging and feature randomness when building each individual tree to try to create an uncorrelated forest of trees whose prediction by committee is more accurate than that of any individual tree.

Logistic Regression Classifier: The Logistic Regression is a classification technique used in machine learning. It uses a logistic function to model the dependent variable. The dependent variable is dichotomous in nature, i.e. there could only be two possible classes (for example: either the cancer is malignant or not).

Support Vector Machine Classifier: SVM is a supervised machine learning algorithm which can be used for classification or regression problems. It performs classification by finding the hyperplane that completely separates the vector

into two non overlapping classes. The vectors that define the hyperplane are the support vectors.

IV. ANALYSIS

A. Confusion Matrix:

A confusion matrix is a table that is often used to describe the performance of a classification model(or "classifier") on a set of test data for which the true values are known. The confusion matrix itself is relatively simple to understand, but the related terminology can be confusing. [8]

There are two labels in a confusion matrix: true label and predicted label. The true label references the actual dataset and the predicted label reference the predicted dataset by the model.

There are two possible predicted classes: "yes" and "no". If we were predicting the presence of a disease, for example, "yes" would mean they have the disease, and "no" would mean they don't have the disease.

The following 4 are the basic terminology which will help us in determining the metrics we are looking for.

- True Positives (TP): when the actual value is Positive and predicted is also Positive.
- True negatives (TN): when the actual value is Negative and prediction is also Negative.
- False positives (FP): When the actual is negative but prediction is Positive. Also known as the Type 1 error
- False negatives (FN): When the actual is Positive but the prediction is Negative. Also known as the Type 2 error.

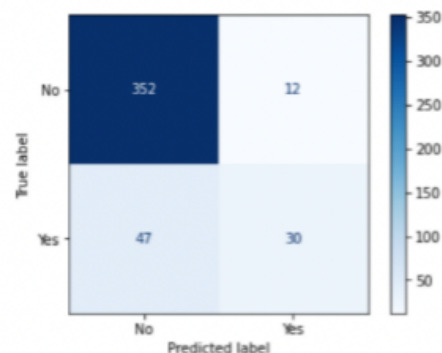


Fig. 3. Confusion matrix figure

TABLE I
LOGISTIC REGRESSION

True Positives	47
True Negative	352
False Positive	12
False Negative	30

TABLE II
RANDOM FOREST MODEL

True Positives	67
True Negative	362
False Positive	2
False Negative	10

TABLE III
SVM MODEL

True Positives	65
True Negative	362
False Positive	2
False Negative	12

B. Classification Measure

Basically, it is an extended version of the confusion matrix. There are measures other than the confusion matrix which can help achieve better understanding and analysis of our model and its performance. Here, we'll analyse our model on the basis of (a.) Accuracy (b.) F1-Score

Accuracy: Accuracy simply measures how often the classifier makes the correct prediction. It's the ratio between the number of correct predictions and the total number of predictions. The accuracy metric is not suited for imbalanced classes. Accuracy has its own disadvantages, for imbalanced data, when the model predicts that each point belongs to the majority class label, the accuracy will be high. But the model is not accurate. It is a measure of correctness that is achieved in true prediction. In simple words, it tells us how many predictions are actually positive out of all the total positive predicted. Accuracy is a valid choice of evaluation for classification problems which are well balanced and not skewed or there is no class imbalance. [9]

TABLE IV
ACCURACY OF OUR MODELS

Model	Accuracy
Logistic Regression	86.394
Random Forest	84.353
SVM	86.394

Here in our paper, we can see that "Logistic regression" and "Support Vector Machine" have the same and the best accuracy.

Recall: It tells us what proportion of Data belonging to a certain class say, class A is classified correctly as in class A by our classifier.

Precision: It tells us what proportion of data that our classifier has classified in a certain class, say class A actually belongs to the same class A.

F1 Score: When the skew in the class distributions are severe, accuracy can become an unreliable measure of model performance. The reason for this unreliability is centered around the average machine learning practitioner and the intuitions for classification accuracy. F1-Score or F-measure

is an evaluation metric for a classification defined as the harmonic mean of precision and recall. It is a statistical measure of the accuracy of a test or model. Mathematically, it is expressed as follows,

$$f1score = (2 \cdot recall \cdot precision) \div (recall + precision) \quad (1)$$

TABLE V
F1 SCORES OF OUR MODELS

Model	F1 Score
Logistic Regression	0.473
Random Forest	0.241
SVM	0.491

As we can see, F1 score of SVM is the highest, followed by logistic regression. Random forest has the least F1 score.

In our paper, we can conclude now that SVM is the best model according to F1 score.

V. RESULT

According to F1 score and Accuracy, the best model out of Logistic Regression, Random Forest and SVM is SVM.

Reasons to choose SVM

- It has the highest F1 score among SVM, Logistic Regression and Random Forest.
- It has highest accuracy among SVM, Logistic Regression and Random Forest.

TABLE VI
RESULT TABLE

Model	Accuracy	F1 Score
Logistic Regression	86.3946%	0.473
Random Forest	86.3946%	0.241
SVM	84.3535%	0.491

VI. CONCLUSION

Employee attrition is not a completely unavoidable problem for organizations. However, they can address it at the right time. Firstly, they need to compute the attrition rate so they can fully understand the issue and the causes behind it. They can then come up with a well-planned, data-driven process to reach out to new job candidates, engage them with different strategies, and retain them for a longer period of time.

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